Proposal: JUST Regulations Assistant

Team Project, Phase 1: Problem and Data · Building LLM Systems: From Models to Production · Jordan University of Science and Technology (JUST), Irbid · 1 Nov 2026

1. Team and roles (Phase 1)

Member	Phase 1 role	Responsible for
Hamza Moeen Rasheed Essa	Product lead + Evaluation lead	User and problem section; the 30 test questions, expected answers and scoring method
Mohammad Ziad Abdelmajeed Al Rousan	Data lead	Collecting the 36 documents with links, checking extracted Arabic text and OCR numbers, the data card
A'sem Zakariya Ali Elfrieh	Engineering lead	GitHub repository, text-extraction pipeline and tests, planned architecture on free compute

Roles rotate at the start of every phase, so each member holds every role by the end of the project.

2. User and problem

Users: JUST students and staff who visit the university website with a rules question. Two concrete groups and tasks: (1) **undergraduate students** checking academic rules: absence limits, withdrawal, exam absence and excuses, grades and GPA, probation, postponement, graduation, discipline; (2) **faculty and administrative staff** checking work rules: office hours, teaching load, administrative and unpaid leave, attendance, summer teaching.

Today: the answers sit in separate official PDFs. People open the documents one by one or ask colleagues and friends, and can get wrong answers on rules that cost grades, fees or leave days.

Evidence so far (from the data itself):

- **Fragmentation:** the official regulations page lists **346 separate PDF documents**, by title only, with no search across their contents. A simple question such as "can I still withdraw?" needs the 17-page degree regulation.
- **Not searchable:** 17 of the 36 documents we selected are scans or use broken fonts, so their text cannot be searched or copied; the other 19 produce **895 corrupted words** when copied (the letter pair "لا" comes out reversed).
- **Conflicting rules:** a student's medical excuse must reach the Health Center within **2 working days** (Medical Excuses Instructions, Art. 4), but the degree regulation gives **2 weeks** to submit it to the dean (Art. 7(e)).

3. What the system will do

A chat assistant for the JUST website that answers questions in Arabic or English, **only from the official documents**, and cites the document and article with every answer. Questions that need a personal record, a committee decision or data not in the documents get "I don't know" plus the office to contact.

> **Student:** لحد أي أسبوع بقدر أنسحب من مادة نظرية؟
> **Answer:** حتى الأسبوع 14 من بداية الفصل (13 للمساقات العملية، وأول 7 أسابيع في الصيفي). يثبت المساق "منسحب" ولا ترصد الرسوم.
> Source: Degree Regulation, Art. 12(b).

> **Staff:** How many office hours do I need per week?
> **Answer:** At least 6 hours a week over at least 4 days (non-clinical faculty). Source: Office Hours Rules, item 6.

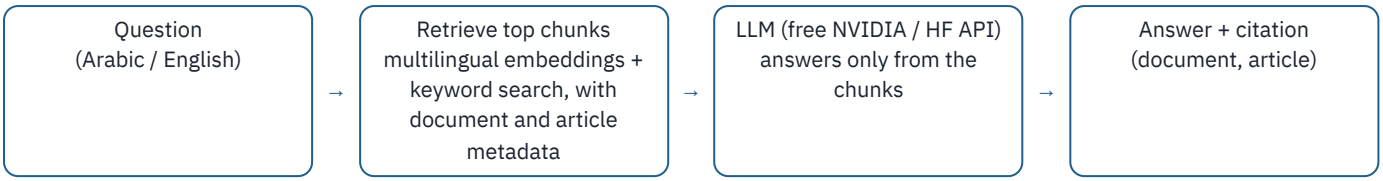
> **Student:** هل رح تقبل اللجنة الطبية عذري؟
> **Answer:** لا أعرف؛ قبول العذر قرار اللجنة الطبية الطلابية في المركز الصحي، ويمكن الاستعلام عن قرارها من خلال عميد كليتك.

4. Data

Source (all on just.edu.jo, official regulations page)	Docs / pages	Status
Student rules: degree regulation, exams, medical excuses, discipline 2024, conduct code, fees, aid and loans, level exams, change of major, insurance, housing, weather suspension, and more	24 / 73	In hand, text extracted
Staff rules: faculty instructions, teaching load, office hours, administrative and unpaid leave, attendance, conferences, summer teaching	12 / 34	In hand, text extracted

36 PDFs, 107 pages, about 247,000 characters of Arabic text (roughly 100k tokens). Details, links and samples: data card and `data/SOURCES.md`.

5. Planned architecture (first version)



If retrieval confidence is low or the question is personal, the system answers "I don't know" and names the office. **Room to grow (RAG + agent):** later phases add calculator tools the agent can call (absence limit for a course, GPA after retaking a course, administrative-leave days from overtime hours), compared against the plain RAG version on the same test set.

6. How we will measure success

- **Answer accuracy** on the test set: the answer matches the expected answer (manual review, with an LLM judge as a helper).
- **Citation accuracy:** share of answers that cite the right document and article.
- **Correct refusals:** share of out-of-scope questions answered with "I don't know" (3 now; more in Phase 2).
- **Retrieval recall@5:** share of questions whose source article is in the top 5 retrieved chunks.

Seed test set: 30 questions (15 easy, 10 medium, 5 hard; 25 student, 5 staff; 21 Arabic, 9 English). Expected answers were written from the documents before running any model.

7. Risks and backup plan

- **OCR errors in numbers and article numbers** (e.g. digits reversed in fee tables). Backup: check every number used in the test set against the PDF page by hand; keep key tables as hand-checked CSV.
- **Conflicting or outdated documents.** Backup: store the document and decision date with each chunk; when two rules differ, give both with their sources.
- **Wrong answers can cost grades, fees or leave.** Backup: always show the source; label the assistant as guidance, not an official decision.
- **Broad scope (students and staff).** Backup: version 1 covers the 36 documents above; more of the 346 are added only after the test set passes.
- **Arabic quality and limits of free models.** Backup: compare models in Phase 2; use a closed model only for development comparison.